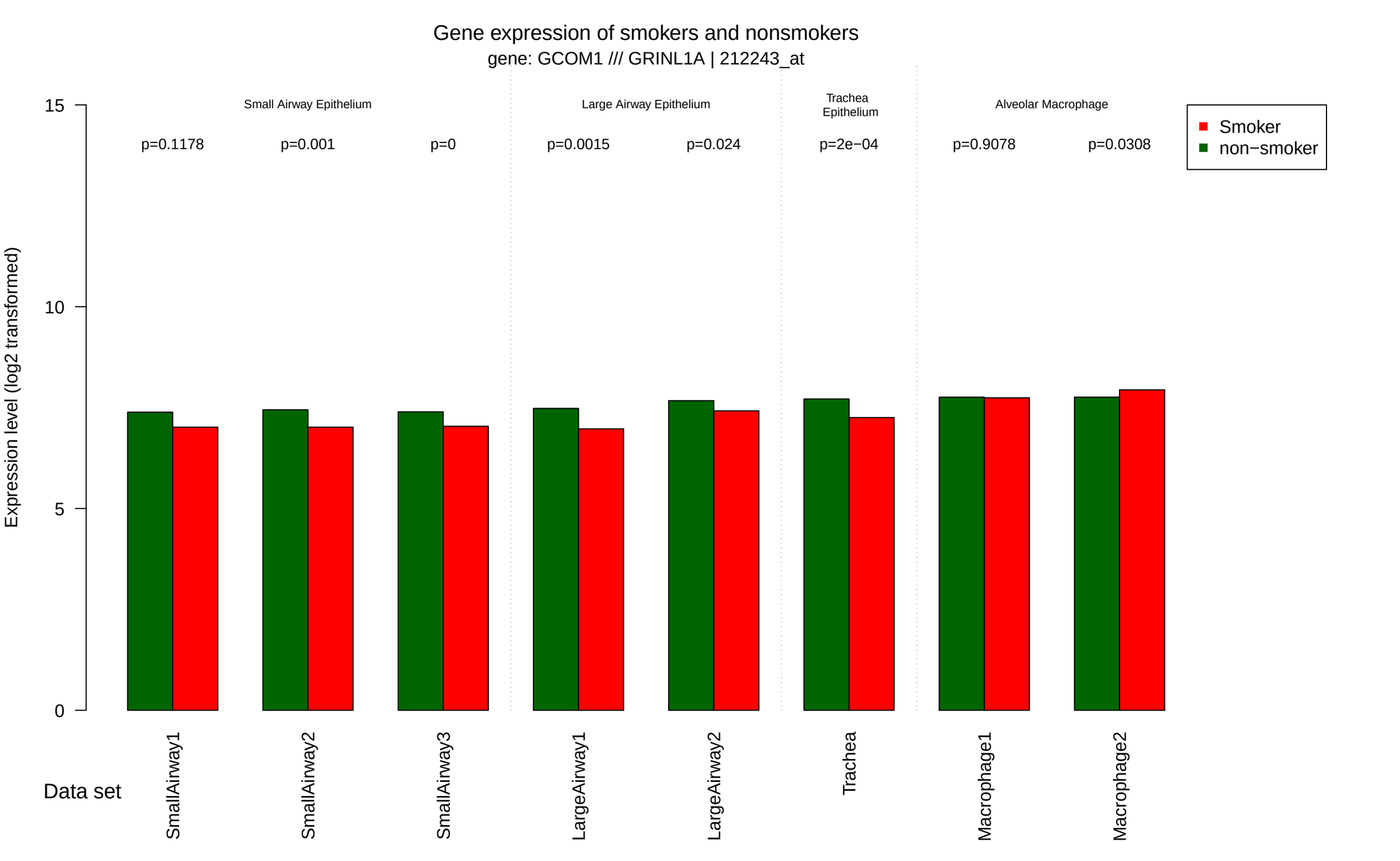
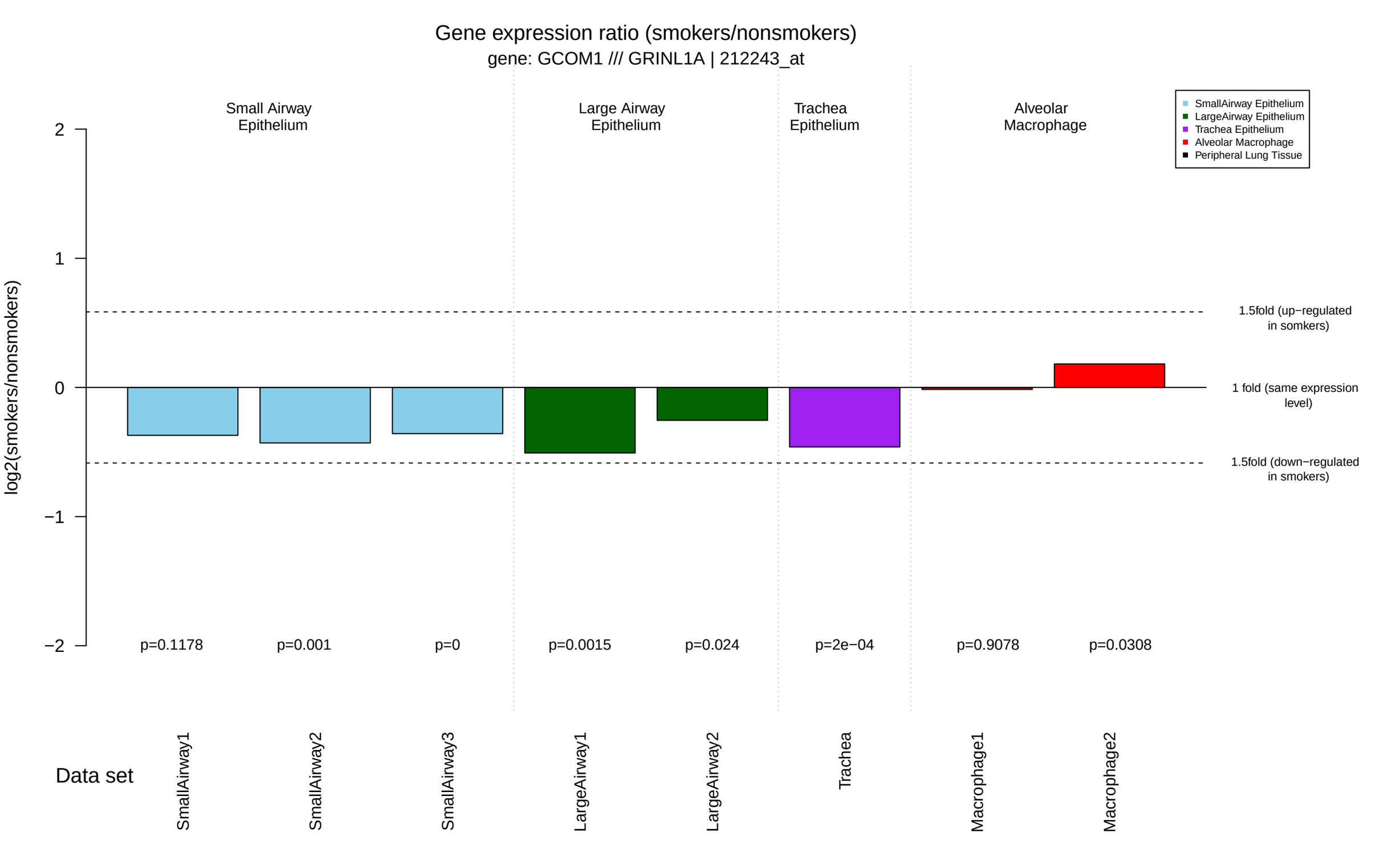
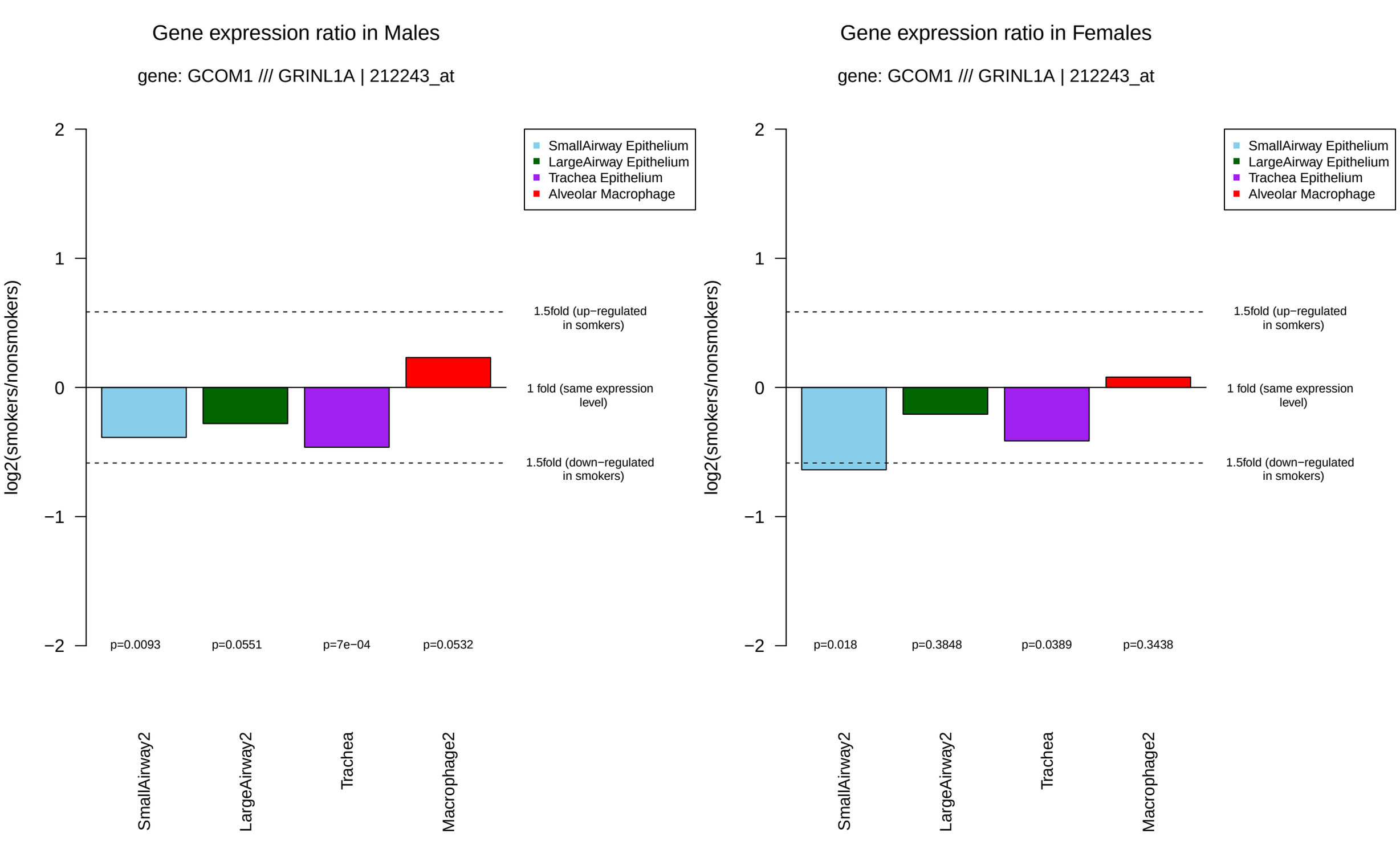
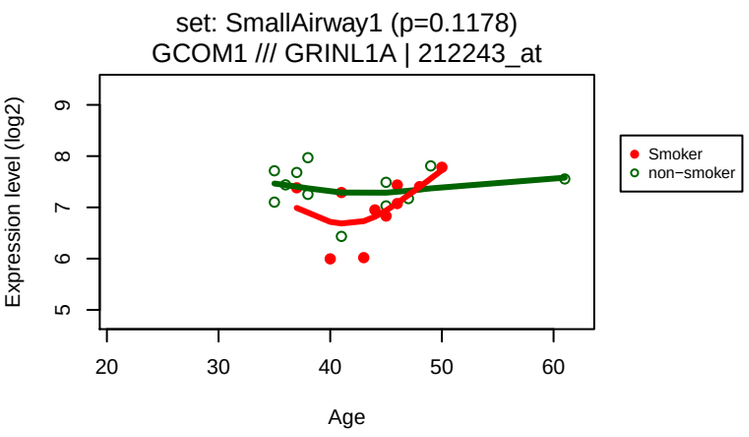
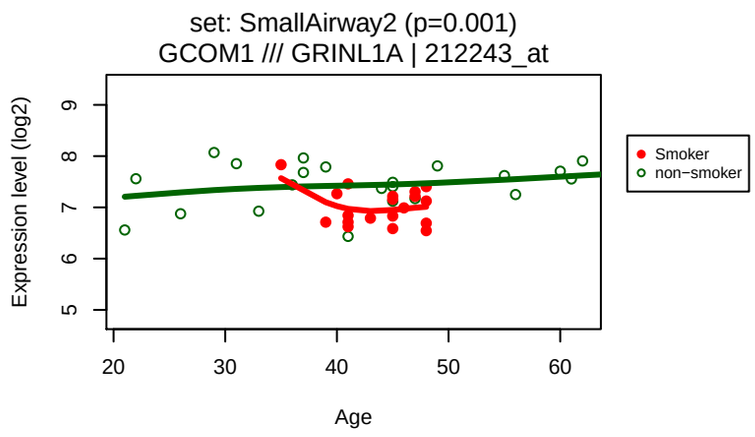
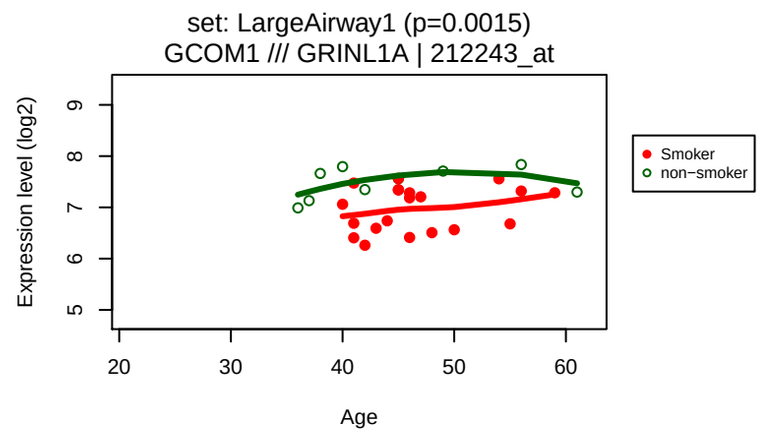

Gene expression of smokers and nonsmokers
gene: GCOM1 /// GRINL1A | 212243_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.1178
p=0.001
p=0
p=0.0015
p=0.024
p=2e−04
p=0.9078
p=0.0308
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

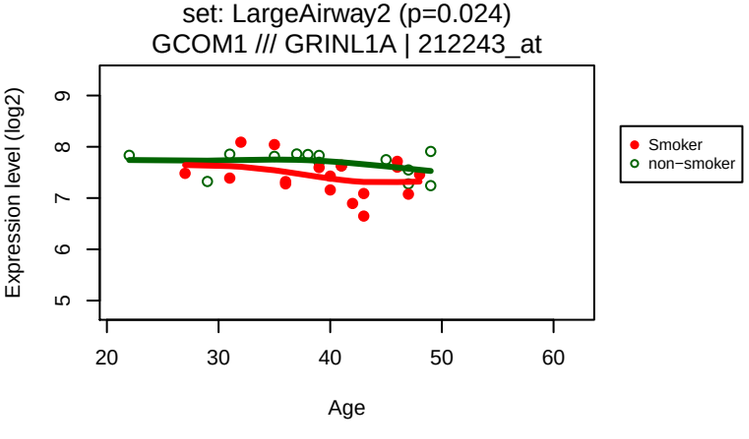

Gene expression ratio (smokers/nonsmokers)
gene: GCOM1 /// GRINL1A | 212243_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.1178
p=0.001
p=0
p=0.0015
p=0.024
p=2e−04
p=0.9078
p=0.0308
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

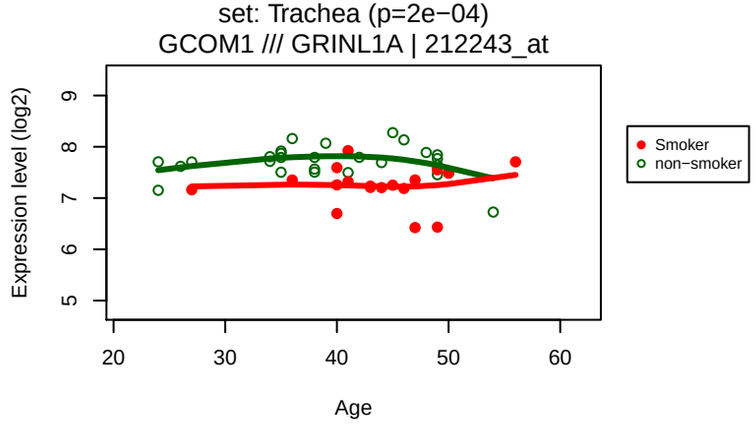

Gene expression ratio in Males
gene: GCOM1 /// GRINL1A | 212243_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.0093
p=0.0551
p=7e−04
p=0.0532
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: GCOM1 /// GRINL1A | 212243_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.018
p=0.3848
p=0.0389
p=0.3438
SmallAirway2
LargeAirway2
Trachea
Macrophage2

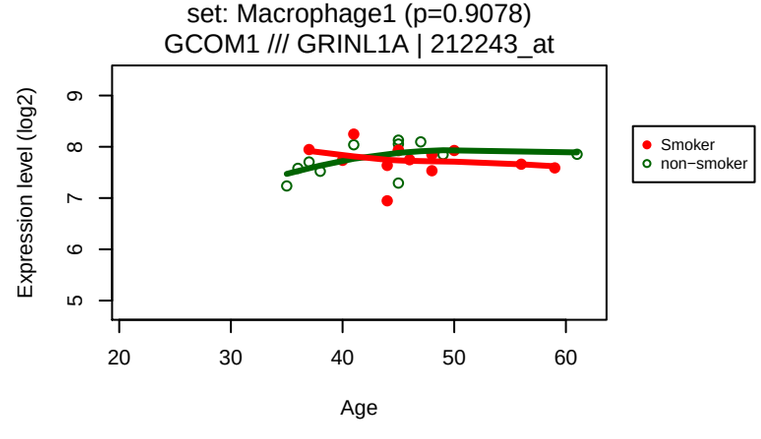

set: SmallAirway1 (p=0.1178)
GCOM1 /// GRINL1A | 212243_at
Expression level (log2)
Age
Smoker
non−smoker

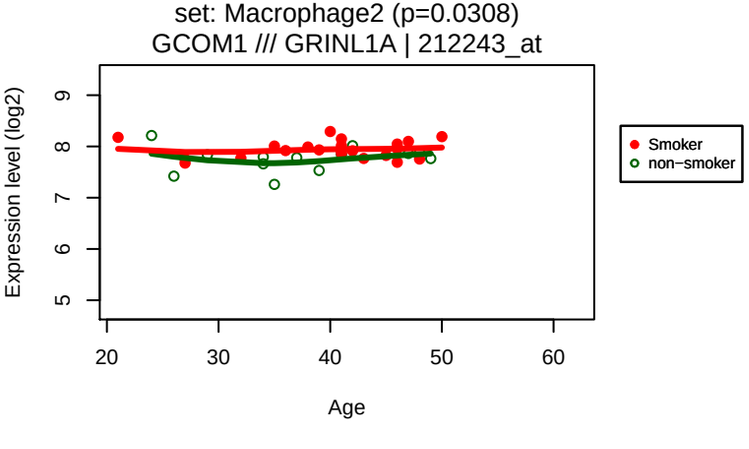

set: SmallAirway2 (p=0.001)
GCOM1 /// GRINL1A | 212243_at
Expression level (log2)
Age
Smoker
non−smoker

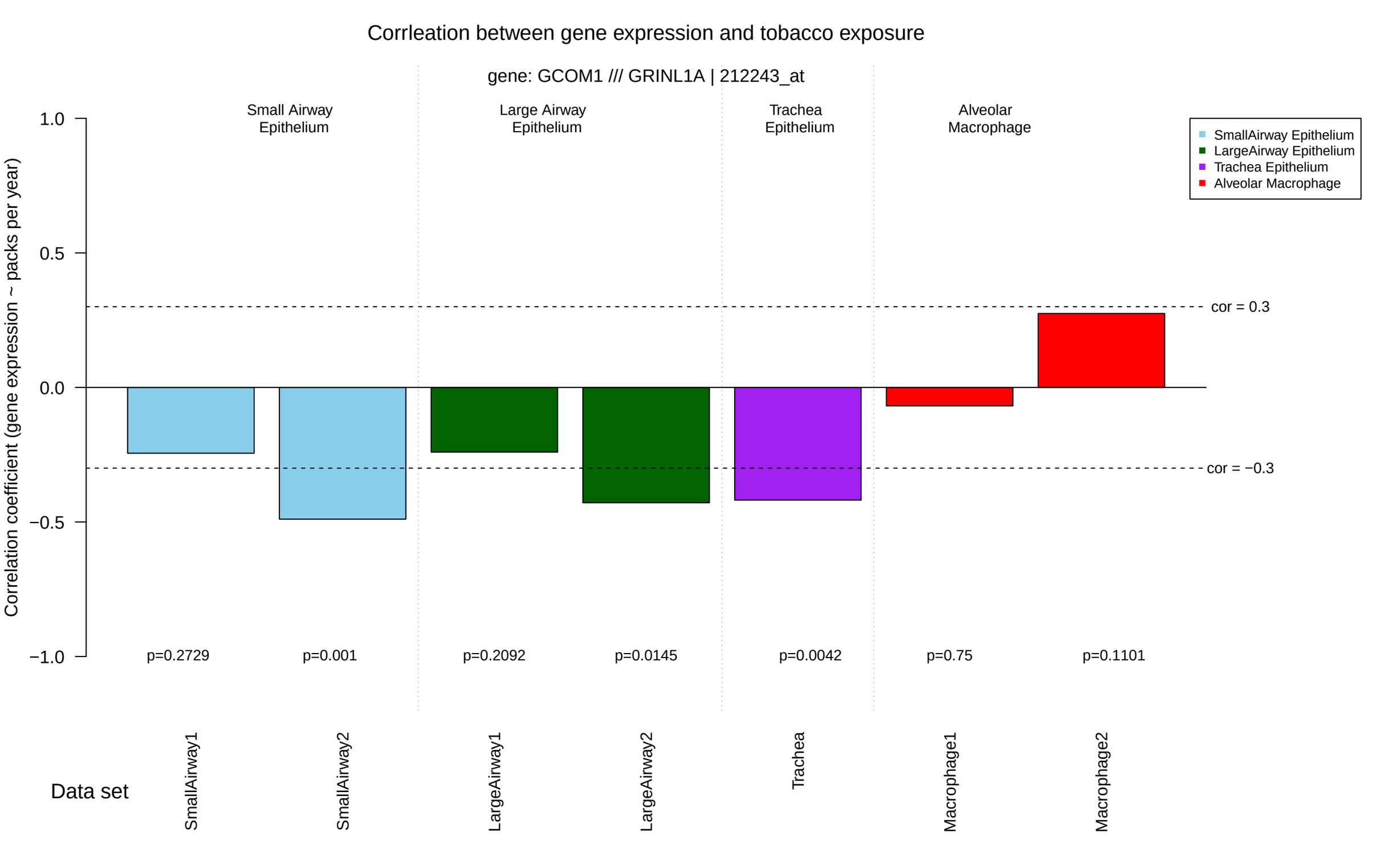

set: LargeAirway1 (p=0.0015)
GCOM1 /// GRINL1A | 212243_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.024)
GCOM1 /// GRINL1A | 212243_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=2e−04)
GCOM1 /// GRINL1A | 212243_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.9078)
GCOM1 /// GRINL1A | 212243_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.0308)
GCOM1 /// GRINL1A | 212243_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: GCOM1 /// GRINL1A | 212243_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.2729
p=0.001
p=0.2092
p=0.0145
p=0.0042
p=0.75
p=0.1101
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2